

CS & IT ENGINEERING



Theory of Computation

- Based CN problems
- Grammar
- PDA
- CFL
- Closure properties

Context
free

(15 - 20)

Language



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DPP-01 Discussion Notes

[MCQ]



#Q. Consider the following grammar:

$S \rightarrow abSba \mid bSb \mid aSa \mid a \mid b \mid \epsilon$

What is the language generated by above grammar?

$\{ \epsilon, a, b, bb, \overline{abba}, \overline{abbbba}, \overline{abbbbaa} \}$

aSa
 $a\epsilon a$
 aa

☒ A CFG that generates all palindromes over alphabet (a, b). ✓

☐ B CFG that generates all palindromes over alphabet (a, b) that do not contain substring "ab".

☐ C CFG that generates all palindromes over alphabet (a, b) that contain substring "aa".

☐ D CFG that generates all palindromes over alphabet (a, b) that do not contain substring "aa".

A

Palindrome:

Order of string does not change while reading from beginning or end.

Ex

$$L = \{ \underline{a}, b, \epsilon, \underline{aa}, \overset{\leftarrow}{\underline{ab}}, \overset{\leftarrow}{\underline{ba}}, \overset{\leftarrow}{\underline{aba}}, \dots \}$$

X X ✓

$$S \rightarrow aSa \mid bSb \mid \cancel{a} \mid \cancel{b} \mid \epsilon$$

asa

abSba

abab

abba

odd length Palindrome

[MCQ]

qsb



#Q. Consider the following grammars:

G1: $S \rightarrow \underline{aSc} \mid \B

$B \rightarrow \underline{bBc} \mid \$$

G2: $S \rightarrow \underline{aaSc} \mid \B

$B \rightarrow \underline{bBc} \mid \$$

G3: $S \rightarrow \underline{aSc} \mid \B

$B \rightarrow \underline{bbBcc} \mid \$$

$$\underline{a^m \$ B c^m} \Rightarrow \underline{a^m \$ b^n \$ c^n c^m}$$

$$\Rightarrow \{ \underline{b^n \$ c^n} \mid n \geq 0 \}$$

$$\{ \underline{a^{2m} \$ b^n \$ c^n c^{2m}} \mid m, n \geq 0 \}$$

abcc

$$L(u_3) \subseteq L(u_1)$$

Which of the following is true regarding G1, G2 and G3?

A

~~$L(G1) \subseteq L(G2), L(G2) \subseteq L(G3)$~~

B

~~$L(G2) \subseteq L(G3), L(G1) \subseteq L(G3)$~~

C

~~$L(G1) \subseteq L(G3), L(G3) \subseteq L(G2)$~~

D

$L(G2) \subseteq L(G1), L(G3) \subseteq L(G1)$

$$S \rightarrow a s b \mid \epsilon$$

$$\underline{a^n b^n \mid n \geq 0}$$

$$L(u_1) = a^m \$ b^n \$ c^h c^m \Rightarrow a^m \$ b^n \$ c^{n+m} \mid m, n \geq 0$$

Push Push Pop
 ↑ skip ↑ ↑

$$L(u_2) \subseteq L(u_1)$$

$$L(u_3) = B = b^{2n} \$ c^{2n}$$

$$\neq \underline{a^{(m)}} \$ \underline{b^{2n}} \$ \underline{c^{2n}} \underline{c^{(m)}}$$

[MCQ]



#Q. Consider the following context free grammar:

$$S \rightarrow \underline{aA} \mid aBB \mid a$$

$$A \rightarrow \underline{aaA} \mid \underline{\epsilon} \mid \underline{aa}$$

$$B \rightarrow bB \mid bbC$$

$$\cancel{C \rightarrow B}$$

①

②

$$\underline{A \rightarrow B}$$

$$B$$

What will be the equivalent simplified CFG for the given grammar?

A

$$S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa \mid b$$

B

$$S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa$$

C

$$S \rightarrow aAa \mid B$$

$$A \rightarrow aaA \mid aa$$

$$B \rightarrow bB \mid bb$$

D

$$S \rightarrow aAa \mid B$$

$$A \rightarrow aA \mid b$$

$$B \rightarrow bB \mid bb \mid a$$

(S, AB, ab)

(1)

$S \rightarrow \underline{aA} / aBB / a$

$A \rightarrow aaA / aa$

$B \rightarrow bB / b\underline{bC}$

$C \rightarrow B$

$\left\{ \begin{array}{l} S \rightarrow aA / a \\ A \rightarrow aaA / aa \end{array} \right\}$

(2)

$S \rightarrow aA / aB / \cancel{aB} / a$

$A \rightarrow aaA / \underline{aa}$

$B \rightarrow \cancel{bB} / b\underline{bB}$

$\begin{array}{ccc} bB & bB & bB \\ bBB & & bB \end{array}$

#Q. Consider the following grammar:

$S \rightarrow AB$

$A \rightarrow BaB \mid a$

$B \rightarrow baA$

AD
 $a b b A \mid a b b B a B$
 $a b b a a a$
 $AB \Rightarrow aB \Rightarrow a b b A$
 $\Rightarrow a b b a$

Which of the following is true regarding given grammar?

A

Every string of the above grammar have at least two a's.

B

Every string have three consecutive a's.

C

Every string have alternate a and b.

D

Every string have b's in multiple of 2.

a, d

[MCQ]



#Q. Given the following two grammars:

G1: $S \rightarrow AB \mid aaB$

$A \rightarrow a \mid Aa$

$B \rightarrow b$

G2: $S \rightarrow aSbS \mid bSaS \mid \epsilon$

What is true regarding above grammars?

A

Both G1 and G2 are ambiguous.

B

Both G1 and G2 are unambiguous.

C

Only G1 is ambiguous.

D

Only G1 is unambiguous.

$amb \Rightarrow UD$

aab

$AmBg$

$gab \Rightarrow asbs$

$abbsas$

$abba$

$asbs$

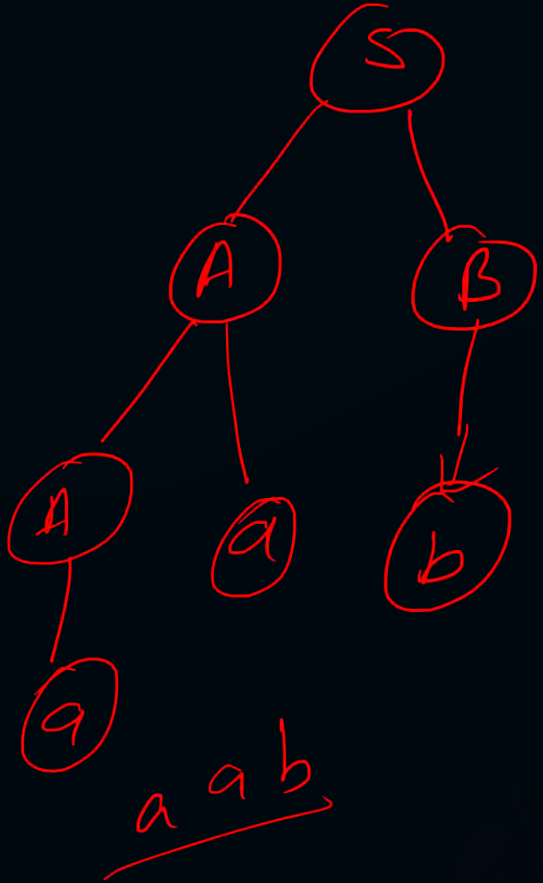
$abbsas$

$abbsac$

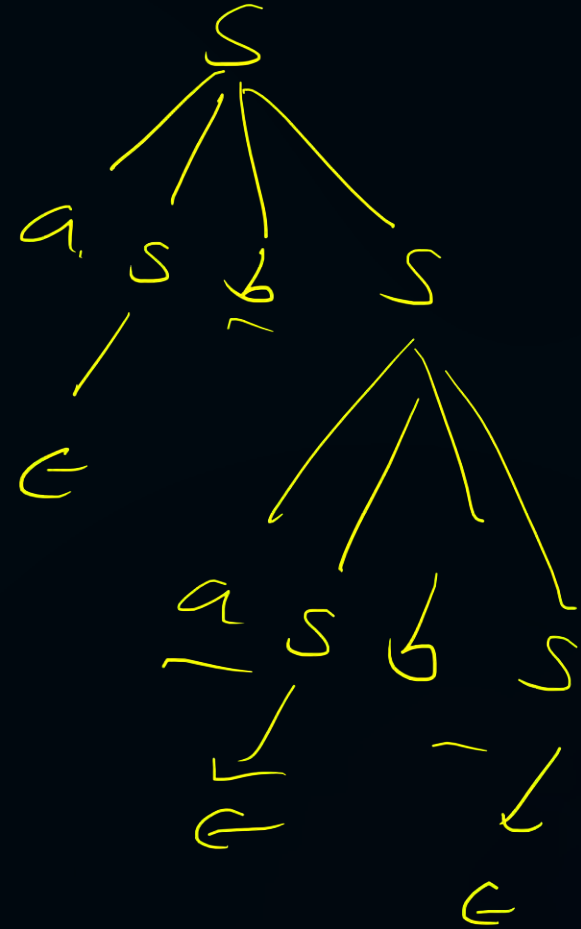
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*If any string having 2 or more parse
tree then that would be ambg*

• aab



abab



[MCQ]



#Q. Consider the following languages

✓ $L1 = \{a^n b^{n+m} c^m \mid n, m \geq 0\}$

Non-Regular, DCFZ, CFL, CSL

✓ $L2 = \{a^p b^q c^{p+q} \mid p, q \geq 0\}$

Non-Regular DCFZ, 1, 1, 1

Which of the following is true?

push push

A

Only L1 is Regular.

X

B

Only L2 is Regular.

X

C

Both L1 and L2 are Regular.

X

D

None of these

✓

#Q. Which of the following is string accepting mechanism of PDA.

A

PDA using final state. ✓

B

PDA using empty stack. ✓

C

PDA using both empty stack and final state. ✓

D

PDA using transition state. ✗

a, b, c

#Q. Which of the following is correct push operation:

A $\Sigma(q, \overset{ab}{\underset{\uparrow}{a}}, \underset{\rightarrow}{b}) = (\underset{\uparrow}{q'}, \underset{\rightarrow}{ab})$ ✓ X

B $\underset{\rightarrow}{\delta}(q, \overset{ab}{\underset{\uparrow}{a}}, \underset{\rightarrow}{b}) = (\underset{\rightarrow}{q}, \underset{\rightarrow}{ab})$ ✓ *self*

C $\underset{\rightarrow}{\delta}(\underset{\rightarrow}{q}, \underset{\rightarrow}{a}, \underset{\rightarrow}{b}) = (\underset{\rightarrow}{q'}, \underset{\rightarrow}{ab})$ ✓

D $\underset{\rightarrow}{\Sigma}(\underset{\rightarrow}{q}, \underset{\rightarrow}{a}, \underset{\rightarrow}{b}) = (\underset{\rightarrow}{q'}, \underset{\rightarrow}{\epsilon})$ X

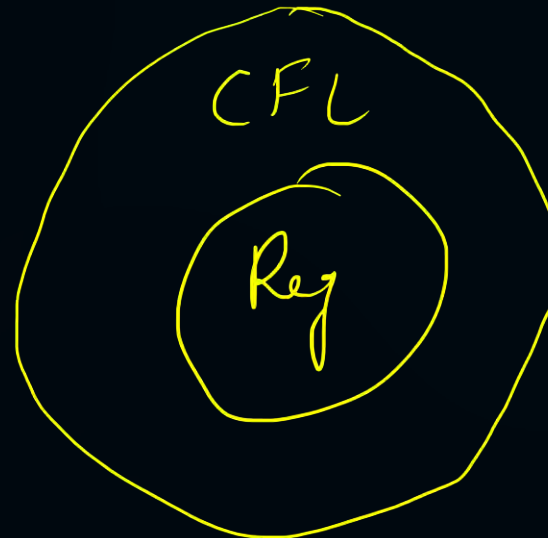
B, C

#Q. Consider the following statements:

- ✓ (I) All DPDA are NPDA.
 - (II) All NPDA are DPDA. ✗
 - (III) All NPDA and DPDA are equivalent. ✗
 - (IV) All context free language are regular language.
- The total number of correct statements are 1.



DCFL $\rightarrow$ CFL
 $\nwarrow$ ✗



[MCQ]



#Q. What does following transition means:

$$\delta(q, \epsilon, b) = (q', b)$$



A

Push b



B

Pop b



C

Read b



D

No operation

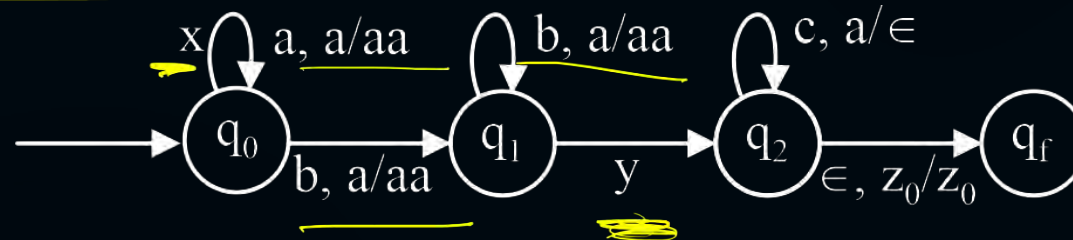


[MCQ]



#Q. What are the values of x and y, if the language accepted by NPDA is

$$L = \{a^m b^n c^{(m+n)} \mid m, n \geq 1\}.$$



A $x = \epsilon, a/a$; $y = c, b/\epsilon$

B $x = a, z_0/z_0$ ~~X~~ ; $y = c, b/c$

C $x = a, a/aa$; $y = b, c/c$

D $x = a, z_0/az_0$; $y = c, a/\epsilon$ ✓

$$a^m b^n c^{m+n}$$

$$a^m b^n c^n c^m$$

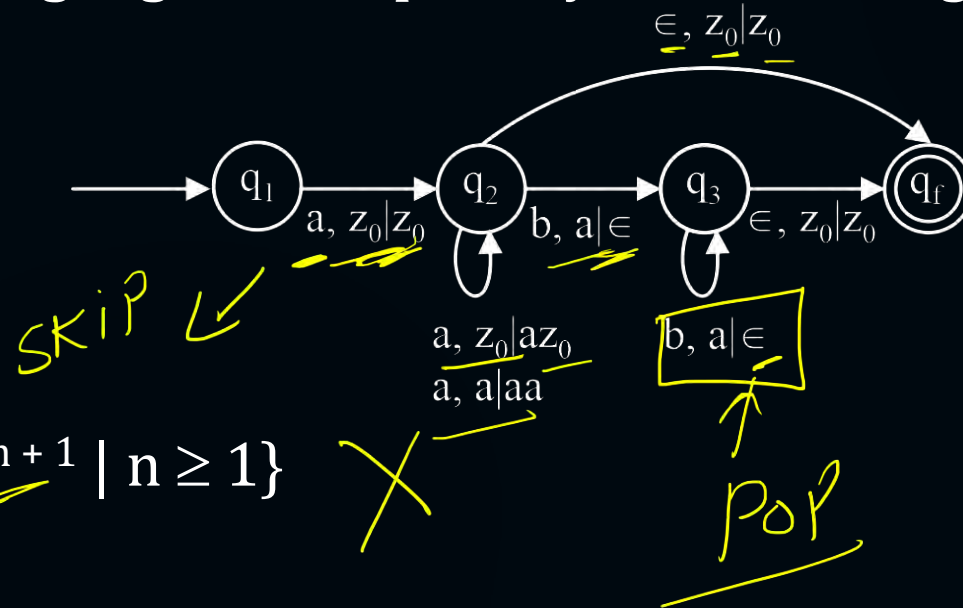
$\uparrow$ $\searrow$
 push $b, a \mid ba$

$$\underline{a, z_0 \mid az_0}$$

$$a^3 b^2 c^2 a^3$$



#Q. Which language is accepted by the following PDA.



$a a^n b^n \mid n \geq 0$
 $n \geq 1$

$a^{n+1} b^n$

A $L = \{a^n b^{n+1} \mid n \geq 1\}$ X

B $L = \{a^{n+1} b^m \mid n, m \geq 1\}$ X

C $L = \{a^{n+1} b^n \mid n \geq 0\}$ ✓

D $L = \{a^n b^{n+1} \mid n \geq 0\}$ X

(E) $\{a^{n+1} b^n \mid n \geq 1\}$

#Q. Which of the following languages are accepted by PDA. CFL

A

$$L = \{a^n b^n c^m \mid m, n \geq 1\}$$

push
↑ ↑ pop skip

$a^n b^n a^+$
CFL

B

$$L = \{a^n b^n c^m \mid m \leq n\}$$

↑ ↑ ↑
 $a^n b^n c^m$ $m \leq n$ X

C

$$L = \{a^n b^m c^n d^m \mid m, n \geq 0\}$$

↑ ↑
↔ ↔

CSL

D

$$L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$$

↑ ↑ ↑
↔ ↔ ↔

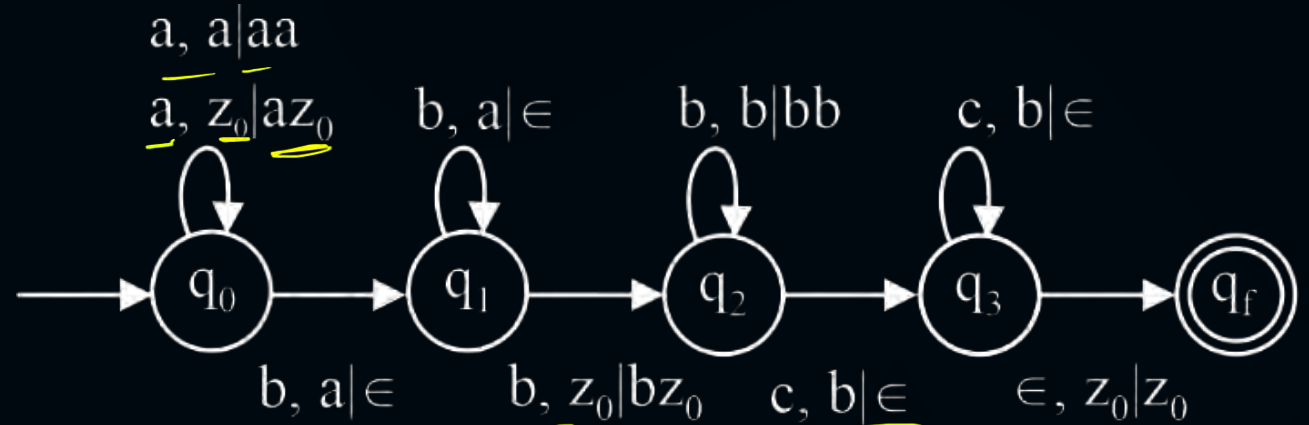
CFL

[MCQ]



#Q. The language derived by given PDA is: *puo^h*

D



A

$$L = \{a^n b^m c^k \mid n = m + k, m, n, k \geq 0\}.$$

B

$$L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 1\}.$$

C

$$L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 0\}.$$

D

$$L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}.$$

Pop

$a^n b^m b^m c^m$

$a^m b^n c^k$

[MCQ]



#Q. Suppose L_1 is a finite language and L_2 is non-regular language then $L_1 \cap L_2$ will be:

$L_1 = \text{Finite} \rightarrow \text{Reg}$

$L_2 = \text{Non-Reg}$

Finite $\cap$ Non-Reg
Inf.

A

Regular but ~~infinite~~

B

Non-regular ~~X~~

C

Finite and regular ✓

D

None of these

[MCQ]



#Q. Consider following statements:

S_1 : Kleene Closure of infinite set always finite.

S_2 : Kleene Closure of finite set always infinite.

Which of the following is correct?

$\{a\}^* = \{\epsilon, a, aa, \dots\}$

false

$\{a\}^* = \epsilon$

A

S_1 only

B

S_2 only

C

Both S_1 and S_2

D

None of these

[MCQ]



#Q. Consider a language L , then subset of L will be?

$$(a+b)^* \Rightarrow \underline{\epsilon}$$
$$\{a^n b^n \mid n \geq 0\}$$

A

Regular



B

Regular but finite



C

Non-Regular



D

None of these



[MCQ]



#Q. Let L_1 be regular language and L_2 be CFL language, then which of the following is correct?

A

A $L_1 \cap L_2$ must be CFL

B $L_1 - L_2$ must be regular

C $L_1 \subset L_2$ must be regular

D None of these

$$\left. \begin{array}{l} L_1 = \text{Reg} \\ L_2 = \text{CFL} \end{array} \right\} \text{Reg} \cap \text{CFL} = \text{CFL}$$

$$L_1 - L_2 \Rightarrow L_1 - L_2$$

$$\Rightarrow L_1 \cap \overline{L_2}$$

$$\text{Reg} \cap \overline{\text{CFL}}$$

$$\text{Reg} \cap \text{CS}^* = \text{CSL}$$

[MCQ]



#Q. Consider two languages L_1 and L_2 .
If $L_1 \subset L_2$ is regular, then which of the following is true?

$$(a+b)^* \Rightarrow C$$
$$L_1 = a^n b^n / n \geq 0$$

L_1 finite / Infinite

A

L_1 must be finite

X

B

L_1 must be regular

X

C

L_1 must be non-regular

X

D

None of these

✓

[MCQ]



#Q. Regular language does not close under on which operation?

(C)

A

Complement ✓

B

Union ✓

C

Subset ✗

D

Intersection ✓



Solve class notes problems

WT, DPP, PYQ, OTS



Good Luck

THANK - YOU